"Concept Note — Var-Rho Noise Discrimination in Xenon TPCs"

Concept Note

Variance–Correlation Discrimination of Correlated Noise in Dual-Phase Xenon TPC Photon Counting

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Document type: concept note (pre-pre-registration). No data are reported. Archived as: https://doi.org/10.5281/zenodo.23499555 (cite this record) Lineage: QCID-Sigma v5.4 → muon pre-registration v1.1 → propulsion-EMI protocol → swarm veto protocol (10.5281/zenodo.23431140) → swarm validation

(10.5281/zenodo.23433610) → front-consistency veto (10.5281/zenodo.23482628) → IceCube DOM study (10.5281/zenodo.23494707) → this note.

1. The problem, in one paragraph
   Dual-phase xenon time-projection chambers — the LZ and XENONnT class of dark matter detectors — watch liquid xenon with photomultiplier arrays sensitive to single photons, and their published background literature reads like a catalog of the exact enemy this program was built to hunt. PMT dark counts that fire together by accident and fake small scintillation signals. Afterpulsing that fakes delayed light. Correlated and delayed secondary scintillation that mimics genuine S1 pulses. Flashing tubes. The single-electron populations that set how low in

energy these detectors can honestly look. These collaborations fight all of this — effectively — with mature, sophisticated pipelines. This note does not propose to replace any of that. It proposes something those pipelines do not contain: a trigger-level noise-rejection layer whose thresholds are frozen before use, whose accepted stream carries certified statistics, and which is cheap and portable enough to run anywhere — a PMT screening bench, a small university-scale TPC, a next-generation sensor program — in units identical to every other lab's.

2. What the gate is, and why it transfers
The mechanism is unchanged from the hardware-verified baseline: a variance stage that rejects the small-amplitude uncorrelated population, a correlation stage that discriminates pulse shape against the

genuine single-scintillation template, operating point Var > 1000, ρ > 0.8, with the hold-off re-derived per rate regime exactly as the IceCube protocol established. Two properties do the work here. The gate is rate-honest — the accepted stream's inter-arrival statistics are certified against the dead-time-corrected null, the property a threshold-stability monitor actually consumes. And it is standardized — an ESP32-class implementation means the same locked criterion set evaluates a phototube batch in one lab and another batch in another lab, and the two answers are comparable without translation.

3. What already exists — and what this note owes it

   This section exists because the first question any xenon physicist will ask is "don't we

already do this?” The honest answer has three parts.

First, the existing rejection layers are real and good, and this note names them rather than waving at them: multiplicity requirements that reject dark-count coincidence clusters by demanding several PMTs fire together; pulse-shape discrimination on the S1 itself, which these collaborations have demonstrated to extraordinary power; afterpulsing and flashing-PMT flags in their calibration and data-quality machinery; and, downstream, analysis-level classifiers on engineered features. A method that ignores these does not get read.

Second, none of these layers certifies what this gate certifies. Multiplicity cuts and PSD accept or reject events; they do not publish a locked criterion set, they do not carry Wilson intervals as a reporting requirement, and they do not demonstrate that the accepted stream

— not the rejected events, the kept ones — obeys the statistics it claims. Their thresholds are tuned on data; this program's are frozen before data. That difference is not a claim of better physics. It is a different deliverable: an auditable standard.

Third — and this is the debt this study must pay in Phase 1 — a quantitative comparison is owed, on identical simulated noise, between this gate and the standard cuts. The planned pre-registration therefore includes a head-to-head arm, with criteria fixed in advance and no cherry-picking: on each injected noise class, Var-Rho rejection must come within a pre-specified margin of the best-performing published cut for that class; below that margin the claim shrinks to the certification deliverable alone. If the gate underperforms the standard cuts on their own ground, the paper will say so and stand on the portability

ground instead. A comparison that can come out either way is the only kind a referee should accept, and this note commits to it in writing.

4. Data reality, stated plainly
   LZ and XENONnT data are not public, and their pipelines are not open to external trigger substitution. So this note defines two phases and does not blur them. Phase 1 is simulation, built entirely from the published instrumentation, calibration, and background literature — PMT models, S1/S2 pulse shapes, dark-count and afterpulsing parameters, the documented correlated-noise populations — with every threshold and the falsifier framework locked before the first simulated event exists. Phase 2, conditional on access, is application to real waveforms with a collaboration's DAQ group or a small-scale open detector, reported only if it materializes.

No data claim is made anywhere in this document.

5. Mapping

   Table 1 carries the argument in one view: every element of the program's hardware-verified baseline has a xenon-TPC counterpart.

   Baseline (QCID / muon / swarm / IceCube)

   Xenon TPC setting

Scintillator/SiPM single-photon pulses

PMT single-photoelectron pulses (S1: tens-of-ns structure)

Correlated common-mode EMI / DOM correlated noise

Dark-count coincidences, afterpulsing trains, delayed-correlated scintillation, flashing PMTs

Discriminator comparator

PMT threshold-level hit stream

Poisson rate-honesty, dead-time-corrected
Threshold-stability certification of the accepted stream

Three layers: pulse / interval / rate
Same, adapted to PMT-array topology

Falsifiers S / F / N
Falsifiers X1–X5, plus the head-to-head arm X6 (§3)

Seed-pooled Run-All notebook
Same discipline

6. Falsifier framework

   Criteria below are proposed defaults; numeric thresholds are locked in the planned pre-

registration before Phase 1 runs. X6’s criteria are locked with the rest — the comparison arm is pre-committed, not decided after the fact.

Table 2. Proposed falsifiers.

ID	Claim protected	Proposed criterion (fixed at pre-registration)
X1	Correlated-noise rejection	≥ 95% rejection of each injected noise class (dark-count coincidence clusters, afterpulsing trains, delayed-correlated emission, flashing-PMT sequences), per class, at the locked operating point
X2	Signal efficiency	≥ 90% retention of simulated genuine S1 and S2 pulses vs the threshold comparator, across the

S1 size range of interest

X3

Rate-honesty

Accepted-stream KS p ≥ 0.05 vs the dead-time-corrected null at all scanned rates; Fano within [0.9, 1.1] at the nominal regime

X4

Threshold portability

Locked point unchanged across PMT-gain spread (published batch variation), rate scan (×10), and noise-mixture variation

X5

No bias

Efficiency uniform across S1 amplitude bins and PMT identity within tolerance; no position-dependent structure beyond it

X6

Head-to-head vs existing cuts

On identical simulated noise, per class: Var-Rho rejection within [M] points of the best published cut for that class, else the claim defaults to certification-only; the paper reports both columns regardless of outcome

7. Status and next step

   This note fixes the mechanism, the comparison debt, and the falsifier framework in advance of any data, in the manner of the program’s six archived records before it. The sequel is a locked pre-registration — X6’s margin M frozen with everything else — then the Phase-1 simulation, archived alongside. Every parameter Phase 1 needs is in the open literature; nothing waits on anyone’s permission. The conversations this work is for

— a small-TPC group, a PMT screening bench, a next-generation sensor program — begin with the locked protocol and the Phase-1 result in hand, not before.

Declarations

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Author contributions: N.K. wrote this note.
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References (indicative)

LZ Collaboration, LUX-ZEPLIN Technical Design Report, arXiv:2002.02764 (to be completed at pre-registration).
LZ Collaboration, commissioning and first-result papers (2022–2024) — trigger/DAQ and background populations (indicative).
XENON Collaboration, XENONnT

commissioning and background literature (2022–2024) (indicative).
LUX Collaboration, single-electron and low-background publications (indicative).
N. Khan, QCID-Sigma v5.4, Zenodo 10.5281/zenodo.23378709.
N. Khan, IceCube DOM noise record, Zenodo 10.5281/zenodo.23494707 — hold-off re-derivation and three-layer precedent.